Event boundaries directionally influence item-level recognition memory

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Abstract

Despite the continuous nature of our experiences, we chunk the moments of our lives into events separated by event boundaries. This event structure affects the way we remember those experiences. Event boundaries have particularly strong influences over the way our memories are temporally organized. However, the way event boundaries influence memory for item-level information is not well understood. Here, we used a mnemonic similarity task with incidental encoding of images blocked into discrete events. In line with prior work, transitions between blocks were operationalized as event boundaries. During a surprise test phase, participants were tasked with distinguishing studied Targets from Novel items as well as similar Lures. Targets and Lures were divided into Non-Boundary items studied in the middle of a given block and Boundary items that either preceded (Pre-Boundary) or followed (Post-Boundary) an event boundary. Target recognition was selectively impaired for Post-Boundary items. In contrast, mnemonic discrimination of similar Lures from studied items was selectively enhanced in Pre-Boundary items. Additional experiments determined that these recognition memory effects depended on an event boundary being characterized not only by a perceptual shift or a task shift, but by a combination of both factors. These findings suggest that event boundaries exert directional effects on recognition memory, promoting high-fidelity encoding of item-level information that immediately preceded these transitions, but leading to poorer recognition for information that immediately followed a transition. Moreover, the influence of event boundaries on memory processes may critically depend on how information is changing to signal an event transition.

Introduction

Our lives unfold as a continuous stream of incoming information. However, we tend to understand and remember our experiences in terms of discrete events (Zacks et al., 2007; Zacks, Tversky, et al., 2001; Zacks & Tversky, 2001). Moments from those events can be meaningfully linked together based on perceptual flow, spatial or situational context, goal states, or a host of other factors (Radvansky, 2012; Radvansky & Zacks, 2014). However, in grouping some moments into an event, we also segment those moments from others. Points in time when segmentation occurs are known as event boundaries, moments when an observer perceives that one meaningful unit of experience has ended and another has begun. We perceive boundaries between complex events at several "grains," where fine-grained segmentation corresponds to simple perceptual changes on the order of a few seconds, and coarse-grained segmentation tends to map onto more general changes in context or goals on the order of many seconds to minutes (Hard et al., 2006; Zacks, Braver, et al., 2001; Zacks, Tversky, et al., 2001). These coarser and less frequent event boundaries tend to capture the way we intuitively perceive the structure of everyday events, and how we store those events in memory (Baldassano et al., 2017).

Event boundaries fundamentally organize human memory by carving meaningful experiences at their joints. Observers show remarkable agreement in where they perceive event boundaries overall (Flores et al., 2017; Sargent et al., 2013; Sasmita & Swallow, 2023). Though there is individual variability, the extent to which a given person agrees with a normative group average predicts how well they will remember details about those events (Flores et al., 2017; Sargent et al., 2013). The act of segmenting events boosts episodic memory for everyday events compared to passive viewing (Richmond & Zacks, 2017). Finally, deficits across cognitive domains - including memory - have associated aberrant event segmentation with aging and Alzheimer's disease (Kurby & Zacks, 2011; Zacks et al., 2006), prefrontal cortical lesions (Zalla et al., 2003), schizophrenia (Zalla et al., 2004), and traumatic brain injury (Zacks et al., 2016). Thus, the way we perceive and learn from the structure of events is a critical area of ongoing research into episodic memory and beyond.

Many studies of event cognition, like those discussed above, test memory using naturalistic stimuli. These often take the form of movies, text stories, or spoken narratives. A challenge in this approach is the difficulty in fully controlling and manipulating experimental variables. As a compromise, a growing number of studies have used simpler stimuli to study event cognition. These studies often present streams of objects as stimuli, and have operationalized event boundaries as changes in perceptual information (Gurguryan et al., 2021; Heusser et al., 2018), context (Ezzyat & Davachi, 2014; Horner et al., 2016), item class (Clewett & Davachi, 2021; DuBrow & Davachi, 2013, 2014, 2016; Sols et al., 2017), reward prediction error (Rouhani et al., 2020), or goal states (Wang & Egner, 2022). A near-universal finding from these types of studies is that memory for temporal order is impaired across event boundaries compared to within-event comparisons. In sum, event boundaries strongly influence memory for "when," leading to a binding of within-event information and a relative separation of between-event information (Clewett et al., 2019; Clewett & Davachi, 2017).

However, we know far less about how event boundaries influence memory for "what." One study by Swallow and colleagues tasked participants with viewing everyday activities, and then tested short-term recognition of perceptual details after a delay of a few seconds (Swallow et al., 2009). Participants showed enhanced recognition of boundary versus non-boundary objects. Interestingly, one experiment from this study tasked participants with distinguishing objects seen in the videos from categorically similar objects. Here, the authors reported evidence that event

boundaries may enhance mnemonic discrimination in addition to basic recognition. This gives rise to three important questions: (1) Do recognition memory enhancements by event boundaries translate over longer delays? (2) How do event boundaries affect the *fidelity* with which specific pieces of information into long-term episodic memory? (3) If such an influence of event boundaries exists, is it directional (i.e., affecting memory for pre-boundary versus post-boundary information), or does it extend bidirectionally?

To answer these questions, we designed a task with an incidental encoding phase with an overt event structure, in line with prior studies. To test the fidelity of recognition memory for items, we used the mnemonic similarity task (MST) (Kirwan & Stark, 2007; Stark et al., 2013, 2019). The MST tests recognition using three types of trials: studied Targets, completely Novel items, and Lures which are perceptually similar but not identical to studied items. The key measures in the MST are correct identification of Targets, which assess basic recognition memory, and correct rejection of similar Lures, which assess mnemonic discrimination.

Mnemonic discrimination is thought to rely on high-fidelity pattern separated representations in the hippocampus (Leutgeb et al., 2007; McClelland et al., 1995; Rolls, 2013; Yassa & Stark, 2011). Indeed, studies using the MST or variants of it have reliably found hippocampal activation during mnemonic discrimination (Bakker et al., 2008; Berron et al., 2016), as well as strong sensitivity to age-related memory decline (Holden & Gilbert, 2012; Reagh et al., 2016, 2018; Stark et al., 2015; Toner et al., 2009). Notably, recent studies have found that the hippocampus is strongly activated at event boundaries, and that this boundary-elicited activity may be related to episodic memory encoding (Baldassano et al., 2017; Ben-Yakov et al., 2013; Ben-Yakov & Henson, 2018; Reagh et al., 2020). Thus, we reasoned that event boundaries might push the hippocampus into a permissive state for encoding episodic memories. In line with this, we would predict that information encoded near an event boundary should be represented with relatively higher fidelity than mid-event information.

We ran three experiments to test this prediction while systematically varying the information contributing to event boundaries at encoding. Experiment 1 defined event boundaries based on a shift in both item class and response type, Experiment 2 defined event boundaries based on a perceptual shift over item class only, and Experiment 3 defined event boundaries based on a task shift in response type only. In the first experiment, we found selective impairments in basic recognition of Targets for items encoded immediately after an event boundary. Conversely, we found selective Lure discrimination enhancements for items encoded immediately before an event boundary. In the second experiment, we show that these effects are not present if the event boundary is defined only by a shift in item class without an accompanying shift in response type. In the third experiment, we show that these effects are not present if the event boundary is defined only by a shift in response type without an accompanying shift in item class.

Results

All experiments shared a common structure overall, with an incidental encoding phase separated by a surprise test phase (Fig. 1). In the encoding phase of Experiment 1, participants viewed alternating blocks of scenes and objects, making “natural or manmade?” judgments for scenes and “indoor or outdoor?” judgments for objects (Fig. 1A). In the encoding phase of Experiment 2, participants viewed alternating blocks of scenes and objects, but made only an “indoor or outdoor?” judgment for each stimulus (Fig. 1B). In the encoding phase of Experiment 3, participants viewed only scenes, but made alternating blocks of “natural or manmade?” and

“indoor or outdoor?” judgments (Fig. 1C). Across all experiments, participants completed a surprise test phase, making “old or new?” judgments in response to each stimulus. Images consisted of studied Targets, Novel items, and Lures that were similar but not identical to studied items (Fig. 1D). In Experiments 1 and 2, we alternated individual object and scene images at test to preclude any sense of event structure. Note that in Experiment 3, given that only scenes were encoded, the test phase consisted only of scenes.

Our study was designed to test the hypothesis that episodic memory encoding is enhanced around event boundaries. Though we were interested in the temporal directionality of these effects, we did not have *a priori* predictions about a particular outcome. As such, we tested whether Target recognition and Lure discrimination at test were enhanced for items encoded *before* an event boundary (Pre-Boundary) or *after* an event boundary (Post-Boundary) compared to items studied mid-event (Non-Boundary). All experiments were run via web browser and faced challenges common to datasets collected online amidst the COVID-19 pandemic. To ensure basic standards of data quality, we applied a fairly liberal set of criteria for excluding participants from analysis (see Materials and Methods for details). Target recognition was calculated via corrected discriminability of Targets from completely novel Foils, or *d’* (z[“old”|Target] - z[“old”|Foil]). Lure discrimination performance was calculated via Lure Discrimination Index, or LDI (Stark et al., 2013, 2019) (p[“new”|Lure] - p[“old”|Foil]) as we recently applied to MST paradigms using old/new responses at test (Delarazan et al., 2023).

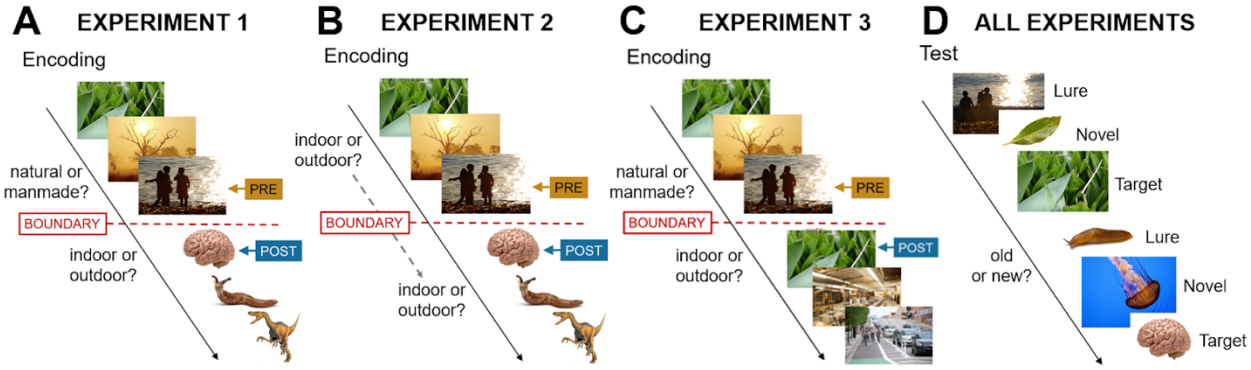

Figure 1: Task schematic. (A) Encoding phase for Experiment 1. Alternating blocks of 10 scenes and 10 objects served as events. Participants made incidental judgments, “natural or manmade?” for scenes and “indoor or outdoor?” for objects. Event boundaries were operationalized as transitions between blocks. (B) Encoding phase for Experiment 2. The task is structured similarly to Experiment 1 with object and scene blocks, except the encoding question did not change. (C) Encoding phase for Experiment 3. The task is structured similarly to Experiment 1 with blocks of alternating judgments, except the stimulus class was limited to scenes and did not change. (D) Test phase, adapted from Mnemonic Similarity Tasks. Participants made “old or new?” judgments for each item. Trials featured studied Targets, Novel items, and similar but not identical Lures. For Experiments 1 and 2, object and scene images were alternated to prevent event structure at test. For Experiment 3, trial types were similar except only scenes were studied and tested.

Experiment 1

Raw correct response rates for Lures, Targets, and Novel items in Experiment 1 (Fig. 1A) are displayed in Table 1 (note that Novel items were not initially encoded, so the “Post, Pre, and Non” designations from the encoding phase do not apply). Overall, participants included in the final sample performed well above chance. Given that stimulus class was manipulated solely to induce event boundaries and not due to *a priori* hypotheses about performance over objects versus scene trials, main analyses combined data across blocks of objects and scenes. Although Lure discrimination was moderately better for objects than scenes, the primary effects we report here

are consistent across stimulus classes (see Supplementary Material for detailed analyses and Supplementary Figs. 1 and 2).

Table 1: Raw response accuracy rates across trial types for Experiment 1. Data are presented as conditional mean alongside standard deviation (SD).

	Object Mean (SD)	Scene Mean (SD)	Combined Mean (SD)
Lure Post CR	0.715 (0.221)	0.616 (0.196)	0.672 (0.159)
Lure Non CR	0.705 (0.131)	0.634 (0.117)	0.673 (0.111)
Lure Pre CR	0.742 (0.209)	0.719 (0.222)	0.730 (0.170)
Target Post Hit	0.717 (0.204)	0.746 (0.186)	0.733 (0.139)
Target Non Hit	0.772 (0.144)	0.783 (0.129)	0.780 (0.112)
Target Pre Hit	0.807 (0.206)	0.7851 (0.196)	0.790 (0.167)
Novel CR	0.949 (0.064)	0.939 (0.069)	0.944 (0.063)

We first wanted to ensure that participants indeed perceived our shifts in experimental conditions as event boundaries. Prior studies have found that responses during encoding tasks slow following an event boundary (DuBrow & Davachi, 2013; Wen & Egner, 2022), which may reflect cognitive costs associated with updating one's event model (Radvansky, 2017; Radvansky & Copeland, 2001; Richmond & Zacks, 2017). Thus, we compared reaction times for items encoded Post-Boundary to Non-Boundary items, as well as items encoded Pre-Boundary. Though there is no reason to suspect any systematic differences between encoded items that would later be designated as Targets or swapped with a similar Lure during the test phase, we also examined Targets and items corresponding to Lures separately to rule out this possibility and its potential influences on recognition performance. In line with our expectations, we found a main effect of item position ($F_{(2,104)} = 63.82$, $p < 0.001$), but no effect of item class (i.e., Lure versus Target, $F_{(1,52)} = 1.75$, $p = 0.191$) and no interaction ($F_{(2,104)} = 0.09$, $p = 0.916$) (Fig. 2A). These differences were driven by slower reaction times for Post-Boundary items compared to Non-Boundary and Pre-Boundary items (all $p < 0.05$ corrected). Thus, participants showed a slower incidental encoding judgment for Post-Boundary trials, indicating that participants indeed experienced event boundaries during encoding.

We then moved on to test our primary hypothesis. Since Non-Boundary trials correspond to typical, well-characterized MST trials (Stark et al., 2013, 2019), these were not of direct interest and served as a baseline comparison for trials adjacent to event boundaries. As such, we used Non-Boundary trials as a baseline, subtracting these from Boundary trials. A negative value would therefore indicate that performance on Pre- or Post-Boundary trials is relatively *worse* than a typical Non-Boundary MST trial, whereas a positive value would indicate *better* performance. Target recognition accuracy was below baseline for Post-Boundary trials ($t_{(52)} = -2.59$, $p = 0.012$). Further, recognition of Post-Boundary targets was significantly worse than Pre-Boundary targets ($t_{(104)} = -2.48$, $p = 0.016$) (Fig. 2B). In contrast, Lure discrimination was above baseline for Pre-Boundary trials ($t_{(52)} = 2.58$, $p = 0.013$), and Pre-Boundary Lure discrimination was significantly better than Post-Boundary Lure discrimination ($t_{(104)} = 2.96$, $p = 0.004$) (Fig. 2C). In sum, we found that event boundaries were *detrimental* to basic recognition of Targets that were studied immediately following the transition, but they *enhanced* discrimination of similar Lures if the original item was studied immediately before the transition.

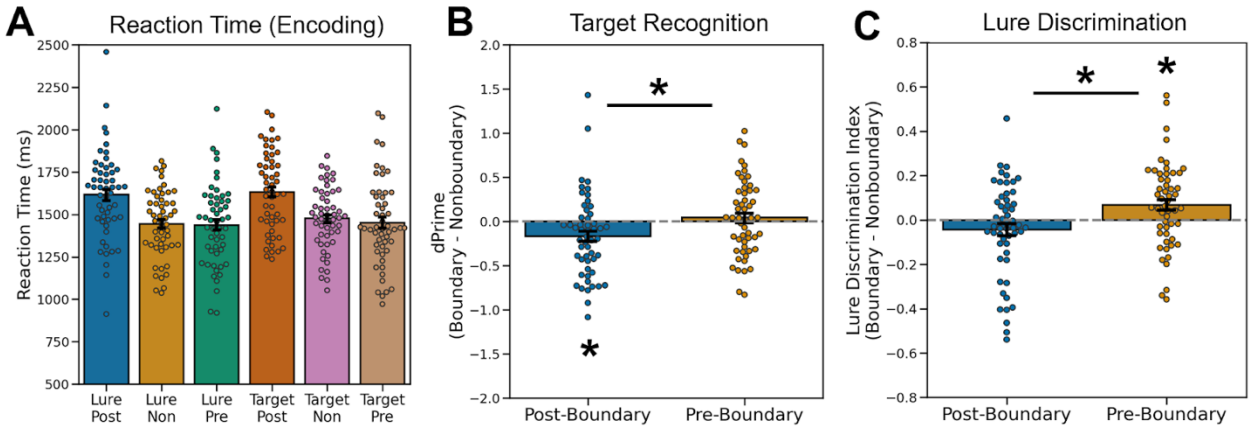

Figure 2: Experiment 1 results. (A) Reaction time data for incidental encoding judgments, broken down by adjacency to an event boundary (Post-Boundary, Non-Boundary, and Pre-Boundary) and corresponding trial type at test (Lure vs. Target). (B) Target recognition performance (d') as a function of adjacency to an event boundary during encoding. (B) Lure discrimination performance (Lure Discrimination Index) as a function of adjacency to an event boundary during encoding. (Asterisks indicate significant result at $p < 0.05$.)

Finally, we note that our encoding blocks were constant in length in this experiment (10 objects or scenes per block). It is therefore possible that the recognition effects reported here result not simply due to event boundaries perceived during encoding, but also growing predictability of the blocked structure of the task over time. To address this possibility, we broke Target recognition and Lure discrimination down into fourths based on when the corresponding studied item was seen during encoding. We reasoned that if trial block predictability was interacting with the perception of event boundaries, then we should see these effects most prominently in test trials corresponding to items studied either in the *first* or *last* quarter of the encoding task. However, this does not appear to be the case for either Target recognition or Lure discrimination (Supplementary Fig. 3).

Experiment 2

Though event boundaries have been the subject of study for decades, they have garnered intense recent interest from episodic memory researchers. Alongside this growing interest, we have seen increasing numbers of tasks designed to study the influence of event boundaries on memory, and a corresponding growth in the number of ways that event boundaries have been operationalized. Importantly, several recent studies suggest that, at least in simplified tasks using streams of static images, shifts in task sets or a participant's goals may be especially important for inducing the perception of event boundaries at encoding (Siqi-Liu & Egner, 2022; Wang & Egner, 2022). As such, we sought to test whether the effects found in Experiment 1 depended upon the nature of experimental shifts operationalized as an event boundary. In Experiment 1, we shifted both the item class (object or scene) as well as the response ("natural or manmade" or "indoor or outdoor?"). For Experiment 2, we defined event boundaries only by shifts in item class, while limiting responses to "indoor or outdoor?" for every stimulus. In other words, this experiment tested the effectiveness of a perceptual-only shift without a change in response in eliciting event boundaries. Data were analyzed in the same way as was described for Experiment 1.

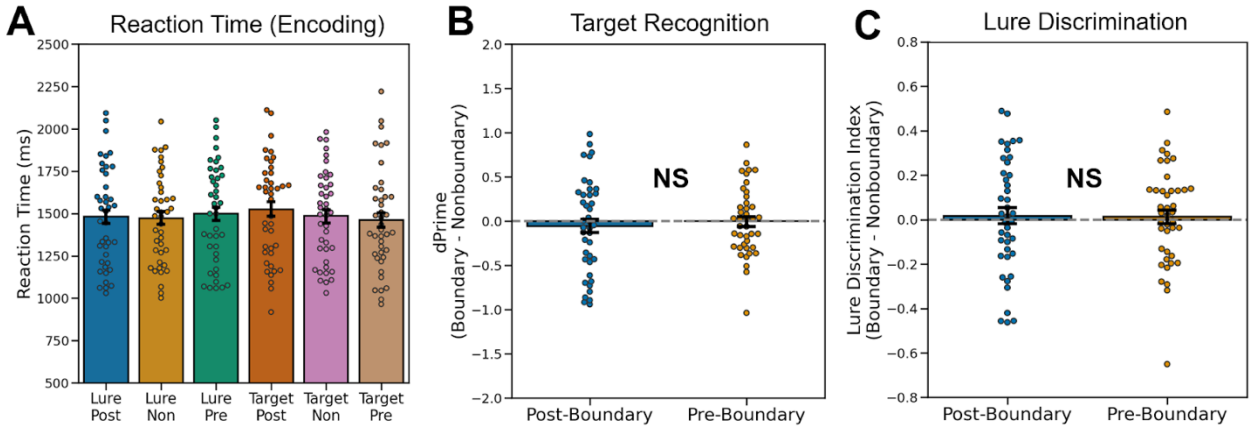

Figure 3: Experiment 2 results. (A) Reaction time data for incidental encoding judgments, broken down by adjacency to an event boundary (Post-Boundary, Non-Boundary, and Pre-Boundary) and corresponding trial type at test (Lure vs. Target). (B) Target recognition performance (d') as a function of adjacency to an event boundary during encoding. (B) Lure discrimination performance (Lure Discrimination Index) as a function of adjacency to an event boundary during encoding. ("NS" indicates nonsignificant statistical comparisons.)

In stark contrast to Experiment 1, we found no modulation of reaction time at encoding based on an item's adjacency to an event boundary (Fig. 3A). Furthermore, we did not find below-baseline Post-Boundary Target recognition ($t_{(40)} = -0.69$, $p = 0.497$) or a difference between Post-Boundary and Pre-Boundary target recognition ($t_{(80)} = -0.72$, $p = 0.474$) (Fig. 3B). Finally, Lure discrimination for Post-Boundary trials was neither above baseline ($t_{(40)} = 0.39$, $p = 0.692$) nor different from Pre-Boundary trials ($t_{(80)} = 0.08$, $p = 0.937$) (Fig. 3C). Thus, in agreement with prior studies, shifts in task sets or goals at encoding appear to be critical for inducing the perception of event boundaries, and in turn producing their effects at retrieval.

Experiment 3

Following from Experiment 2, we next conducted an experiment using the inverse procedure. That is, for Experiment 3 we varied the encoding question across trials ("natural or manmade" or "indoor or outdoor?") while holding item class constant, limited only to scenes. This experiment tested the effectiveness of a response-only shift without a change in item class in eliciting event boundaries and producing the recognition effects reported in Experiment 1. Data were analyzed in the same way as was described for Experiments 1 and 2.

For reaction time analyses, we found a main effect of item position ($F_{(2,100)} = 46.85$, $p < 0.001$), but no effect of item class (i.e., Lure versus Target, $F_{(1,50)} = 0.56$, $p = 0.455$) and no interaction ($F_{(2,104)} = 1.95$, $p = 0.147$) (Fig. 4A). As with Experiment 1, these differences were driven by slower reaction times for Post-Boundary items compared to Non-Boundary and Pre-Boundary items (all $p < 0.05$ corrected), indicating the perception of event boundaries during encoding. However, unlike Experiment 1, we found only a nonsignificant trend toward below-baseline Target recognition ($t_{(50)} = 1.76$, $p = 0.085$) and no difference between Post-Boundary and Pre-Boundary recognition ($t_{(100)} = 0.58$, $p = 0.564$) (Fig. 4B). Similarly, Pre-Boundary Lure discrimination was neither above baseline ($t_{(50)} = 0.98$, $p = 0.331$) nor different from Post-Boundary lure discrimination ($t_{(100)} = 1.66$, $p = 0.104$) (Fig. 4C). In sum, although shifts in response alone in Experiment 3 seem to have elicited event boundaries at encoding (unlike shifts in item class alone as in Experiment 2), they did not reproduce the recognition memory effects observed in Experiment 1. The influence of event boundaries on recognition memory may therefore depend on salient event

transitions signaled by multiple sources of information. Furthermore, alongside Experiment 2, these findings suggest that different sources of information may provide relatively stronger or weaker perceived event transitions.

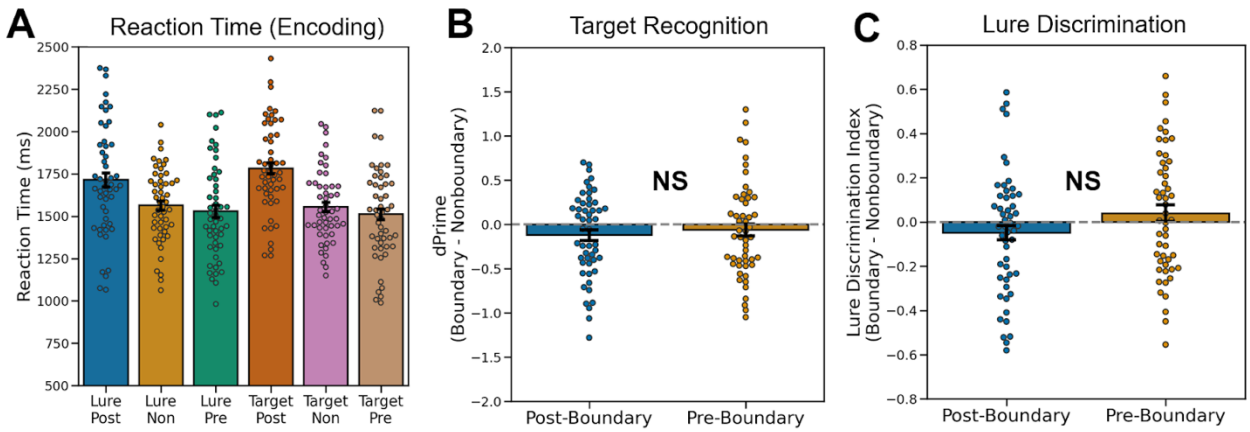

Figure 4: Experiment 3 results. (A) Reaction time data for incidental encoding judgments, broken down by adjacency to an event boundary (Post-Boundary, Non-Boundary, and Pre-Boundary) and corresponding trial type at test (Lure vs. Target). (B) Target recognition performance (d') as a function of adjacency to an event boundary during encoding. (B) Lure discrimination performance (Lure Discrimination Index) as a function of adjacency to an event boundary during encoding. ("NS" indicates nonsignificant statistical comparisons.)

Discussion

This study tested the prediction that event boundaries selectively enhance encoding of item-level information, leading to enhancements in detailed recognition memory. Across three experiments, we adapted a common approach in episodic memory studies using simple blocks of static images, with transitions between blocks operationalized as event boundaries. Consistent with our predictions, we found that event boundaries enhanced Lure discrimination when item memory was tested (Experiment 1). However, this enhancement was directional, applying to items encoded *before* but not *after* an event boundary. Conversely, we found relatively poorer basic Target recognition for items encoded *after* an event boundary. Subsequent experiments showed that these memory effects depended on multiple sources of information informing participants that an event boundary occurred. That is, neither shifts in item class (Experiment 2) nor in the encoding response (Experiment 3) alone were sufficient to elicit these Lure discrimination and Target recognition effects.

A growing body of literature suggests that event boundaries are important elements of experience that can profoundly impact episodic memory (Kurby & Zacks, 2008; Zacks, 2020; Zacks, Tversky, et al., 2001). Prior studies have most reliably shown a disruption of temporal memory for between-event comparisons compared to within-event comparisons, suggesting that event boundaries lead to a binding of information within an event (Clewett et al., 2019; Clewett & Davachi, 2017; DuBrow & Davachi, 2013, 2014, 2016). However, far fewer studies have investigated how event boundaries influence memory *for the items comprising an event themselves*. A prior study by Swallow and colleagues found a relative enhancement of recognition memory for objects encoded near event boundaries after brief delays of a few seconds (Swallow et al., 2009). Here, we build on the existing literature in showing that event boundaries indeed influence recognition memory for items, but that these effects in long-term memory are directional, and differ based on the fidelity of the representation being tested. That is, we found a selective enhancement of Pre-Boundary

Lure discrimination but a relative diminution of Post-Boundary Target recognition. This asymmetry adds important nuance to the idea that event boundaries simply benefit recognition memory.

What might this asymmetry mean? Whereas cortical networks involved in event memory seem to represent the middle of an event or even its entire duration, hippocampal representations appear to be selectively engaged during event transitions (Reagh & Ranganath, 2023). Prior studies suggest that encoding-related hippocampal activity peaks at event boundaries (Baldassano et al., 2017; Ben-Yakov & Henson, 2018; Reagh et al., 2020), which may signal preferential episodic encoding of the event that has just ended (Ben-Yakov & Dudai, 2011; Lu et al., 2022; Silva et al., 2019). In the context of the present data, event boundaries may have driven the hippocampus to take a retroactive "snapshot" of the previous event. In doing so, the representation of the Pre-Boundary event immediately preceding that event's offset may have been strongly captured in this "snapshot," leading to a high-fidelity representation that is conducive to pattern separation (Yassa & Stark, 2011) and resistant to interference from Lures at test. Thus, one side of this phenomenon is a selective memory enhancement for fine-grained information that underlies mnemonic discrimination.

In the other direction, somewhat surprisingly, we observed poorer basic Target recognition after event boundaries. The mechanism of this effect may be akin to the attentional blink (Dux & Marois, 2009; Shapiro et al., 1997), whereby processing of two target stimuli in rapid succession leads to frequent failures to detect the second target. Though the stimuli here were not presented rapidly and no such target detection task was used, it is possible that having to update one's event model at an event boundary (Radvansky, 2012; Richmond & Zacks, 2017; Zacks & Tversky, 2001) leads to a relative "blind spot" for information present immediately at the onset of the new event. Interestingly, the aforementioned study by Swallow and colleagues found that recognition of boundary-adjacent objects may in some cases be worse if the object being tested was from the *current* event and not the *previous* event (Swallow et al., 2009). Taking all the evidence together, we speculate that this asymmetry may arise from episodic encoding resources being devoted to the prior event (Lu et al., 2022) at the expense of the new event. Such an effect, however, likely depends on the perceived importance or meaningfulness of the first information encountered at event onset.

Our results carry several other implications. While a task such as the MST is sensitive to high-fidelity item encoding as supported by the hippocampus (Bakker et al., 2008; Kirwan & Stark, 2007; Stark et al., 2013, 2015, 2019; Yassa & Stark, 2011), these item-level representations can be significantly influenced by event structure. To an extent, understanding an *event* means understanding item information in relation to the context in which it occurs (Reagh & Ranganath, 2018). Thus, we *should* expect event structure to influence the way items are encoded and remembered. These experiments provide such evidence for the influence of event structure on item memory as tested by the MST. Additionally, in the present data, not all shifts in experience truly create the experience of event boundaries, nor are all things we might describe as event boundaries equal in influencing memory. Perceptual shifts in the category of items alone were insufficient even to evoke reaction time effects that characterize event boundaries (Experiment 2). In agreement with prior studies, shifts in task sets seemed to elicit stronger event boundaries (Siqi-Liu & Egner, 2022; Wang & Egner, 2022). This aligns with recent data suggesting that hippocampal encoding mechanisms may be especially sensitive to "cognitive contexts" such as varying rule sets during an encoding task (Dimsdale-Zucker et al., 2022). However, in our data, encoding task shifts alone were not sufficient to produce the memory effects observed when these shifts also coincided with shifts in item category (Experiment 3). Though prior studies have examined the role of different sources of information in producing event boundaries (Kurby & Zacks, 2012), growing interest in this topic calls for further study.

There are several limitations of note across these experiments. First, as described previously, we did not vary encoding block length. Given the number of factors that could potentially influence the encoding and subsequent recognition of items in the paradigm, this was done in part to hold temporal intervals constant and eliminate a possible source of variance. However, an unintended consequence of this may be that the structure of encoding blocks became predictable to participants, and this predictability may have interacted with the way we operationalized event boundaries per se. Though exploratory analyses suggest that this is not likely the case (Supplementary Fig. 3), future studies with variable block lengths can address the issue of the predictability of event durations more directly. Second, as with many prior memory experiments, we operationalized event boundaries in a stripped-down and simplified way compared to real-world experiences. Experiments in line with the study by Swallow and colleagues (Swallow et al., 2009) incorporating long-term memory tests can further bridge this gap. Third, our predictions and conclusions here are at least somewhat motivated by hippocampal mechanisms related to event cognition and high-fidelity episodic memory encoding. Simply put, the study arose from the prediction that the hippocampus' preferential activation at event boundaries amplifies novel encoding, which in turn creates a permissive state to enhance pattern separation (Yassa & Stark, 2011). Though the experiments here did not rely on neural evidence, future investigations incorporating such evidence will provide further clarity about the mechanisms at play.

In sum, this study provides new insights into the influence of event structure on episodic memory. We show that event boundaries exert directional influences on recognition, driving enhanced mnemonic discrimination for Pre-Boundary information but poorer basic recognition of Post-Boundary information. We further show that these effects and the extent to which an event boundary is perceived depends on the information sources signaling a shift in situation. This complements prior work establishing the influence of event boundaries on memory for "when," and builds toward a fuller understanding of their influence on memory for "what."

Materials and Methods

Participants

Participants were recruited from the Washington University undergraduate community and were compensated with course credit. All participants gave written informed consent in accordance with the Washington University Institutional Review Board. Participants were required to be fluent English speakers free of major psychiatric or neurological disorders. Experiments were run online, and participants were required to have a stable internet connection. Power analyses suggested a final sample size of 44 participants was needed. We recruited more participants than necessary to account for attrition due to the online testing format. For **Experiments 1, 2, and 3**, we recruited 100, 78, and 92 participants respectively (exact recruitment numbers were largely based on availability from the undergraduate participant pool). We implemented two exclusion criteria: (1) button responses to fewer than 80% of trials, and (2) raw Target hit and Lure rejection rates numerically below chance performance of 50% accuracy. We note that these are liberal exclusion criteria, and that such choices were made to mitigate data quality issues arising from online data collection during the COVID-19 pandemic. Critically, the criteria applied here were agnostic to experimental manipulations or task conditions. Our exclusion criteria brought data in line with expected response accuracies for healthy young adults in prior studies (Stark et al., 2013, 2019). However, in line with pilot experiments, we excluded a greater proportion of participants compared to what we would expect from in-person samples. After exclusions, **Experiment 1** retained 53 participants (44 female, age = 20.06 ± 1.29), **Experiment 2** retained 41 participants (27 female, age = 19.56 ± 1.18), and **Experiment 3** retained 51 participants (33 female, age = 19.75 ± 1.23).

Materials, Design, and Procedure

Stimuli consisted of images of common objects and scenes curated by Craig Stark and colleagues for conducting experiments using the MST (Stark et al., 2019) (link to stimuli: https://faculty.sites.uci.edu/starklab/mnemonic-similarity-task-mst/). All experiments were conducted online via web browser (required to be either Google Chrome or Mozilla Firefox). Tasks were programmed in JavaScript using the jsPsych package (de Leeuw, 2015) and were hosted on servers maintained by Cognition.run (https://www.cognition.run).

Across all experiments, participants first completed an encoding phase and subsequently completed a surprise test phase. In **Experiment 1**, encoding consisted of 12 blocks of 10 object stimuli alternating with 12 blocks of 10 scene stimuli. Transitions between blocks were operationalized as event boundaries. Thus, there were 24 event boundaries, and 240 images in total. Stimuli were presented for 3 seconds, with a 0.5 second inter-stimulus interval (ISI). Participants were instructed to answer an "indoor or outdoor?" encoding question in response to objects and "natural or manmade?" in response to scenes, using the "F" and "J" keys respectively. Stimulus order was pseudo-randomized, ensuring that the first block encountered was a scene block for half of the participants. During the test phase, participants viewed a randomized ordering of alternating object and scene images to ensure that no event structure was perceived. Participants viewed 360 images, evenly divided into 120 previously studied Targets, 120 completely Novel items, and 120 similar but not identical Lures. Half of all stimuli (e.g., 60 of the 120 Targets) were objects and the other half were scenes. Stimuli were presented for 3 seconds with a 0.5 second ISI. For each test trial, participants were instructed to respond "old" with the "F" key and "new" with the "J" key to the stimulus on the screen. Importantly, participants were told to respond "old" only when they believed that an image on the screen was previously seen before, and that a "new" response should be given for completely new Novel images as well as Lure images that were similar but not identical to something seen in the encoding phase. **Experiment 2** was identical to Experiment 1 in timing and format, with the only difference being that the response given at encoding did not differ as a function of item class. That is, participants saw 12 blocks of 10 object stimuli alternating with 12 blocks of 10 scene stimuli but responded "indoor or outdoor?" in response to each stimulus. The test phase did not differ. **Experiment 3** featured 12 blocks of 10 scenes and featured no object images. These blocks were characterized by the encoding question alternating between "indoor or outdoor?" and "natural or manmade?" every 10 trials. This resulted in 120 trials in total, half of the total trials in Experiments 1 and 2 simply due to the smaller number of experimentally normed scene stimuli available. The test phase featured only scene trials, which alternated based on the encoding question. Participants viewed 180 images, evenly divided into 60 Targets, 60 Novel images, and 60 Lures. Experiment 3 was otherwise the same as Experiments 1 and 2 in timing and format.

Analyses

There were three main conditions of interest based on event structure at encoding. These were: (1) Non-Boundary items, which were viewed in the middle of a stimulus block, (2) Post-Boundary items, which were viewed immediately after an event boundary, and (3) Pre-Boundary items, which were viewed immediately before an event boundary. Exploratory analyses of pilot studies suggested that recognition memory effects reported in the Results section did not extend past items adjacent to event boundaries. For example, Lure discrimination enhancements were present for the item immediately preceding an event boundary (item N-1), but this effect did not extend to the item *before* a Pre-Boundary item (item N-2).

Reaction time data at encoding were averaged across participants as a function of the three conditions described above: Non-Boundary, Post-Boundary, and Pre-Boundary. Though there was no reason to suspect that an item later being designated as a Target or a Lure at test would have any influence on the perception of event boundaries at encoding, we nonetheless split trials as such in these analyses to rule out such a confound. Data were compared via 2-way ANOVA (boundary condition by test trial type), and pairwise contrasts were conducted via Tukey's HSD.

For test data, we analyzed Target recognition and Lure discrimination. Target recognition was calculated as *d'* (z["old"|Target] - z["old"|Foil]), a Signal Detection Theory metric that adjusts raw recognition rates for response bias. Lure discrimination was calculated via Lure Discrimination Index (Stark et al., 2013, 2019), or LDI (p["new"|Lure] - p["old"|Foil]) as applied to MST paradigms using old/new responses at test (Delarazan et al., 2023). These bias-corrected metrics are used for our main analyses, but raw response rates are shown in Table 1. Further, our primary analyses for Experiments 1 and 2 combined data across object and scene blocks given that we had no *a priori* reason to assume that our effects of interest would depend on stimulus class. Nonetheless, we analyzed data for these stimuli separately as reported in Supplementary Figures 1 and 2.

Test data were analyzed with a focus on an item's position at encoding. We were primarily interested in (1) how Post-Boundary and Pre-Boundary trials would differ from Non-Boundary trials, and further (2) in how Post-Boundary and Pre-Boundary trials would differ from each other. We therefore analyzed *d'* and LDI results in two ways. First, we baselined Post-Boundary and Pre-Boundary trials against Non-Boundary trials and ran one-sample t-tests comparing the trials of interest against this baseline. Second, we compared Post-Boundary trials against Pre-Boundary trials using independent-samples t-tests.

As noted in the Results section, encoding blocks all featured10 items per block. Though constant block length was used to minimize the number of variables that could influence boundary-related effects on recognition, this may potentially have led to a confound of block predictability as participants gained experience with the task. That is, participants' growing ability to predict impending transitions between encoding blocks may have exerted an influence that interacted with (or even masked) the effect of event boundaries per se. We ran an analysis to explore this possibility, breaking *d'* and LDI measures down as a function of when the item was seen during encoding. If predictability of the task blocks interacted with or drove the reported effects, then we reasoned that the effects should be seen only for stimuli encountered *either* early or late in encoding. We broke Test data into temporal quartiles of the encoding task, and repeated the main analyses described above. As we show in Supplementary Figure 3, the evidence is not consistent with a major influence of participants simply learning to predict when the event boundaries would occur. That is, the effects were not obviously driven by trials seen either early or late in encoding.

Data and Code Availability

Code for running the experiment, anonymized data files, and analysis scripts are available on GitHub at the following link: https://github.com/complexmemlab/MST_Events_2023.

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Supplementary Materials for:

Event boundaries directionally influence item-level recognition memory

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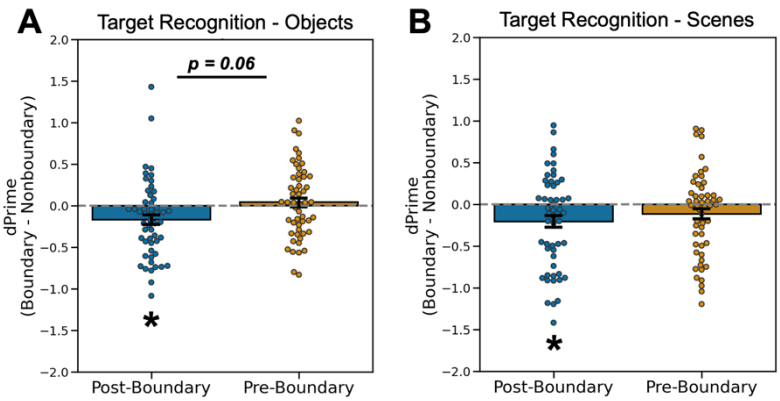
Experiment 1 analyses broken down by objects and scenes

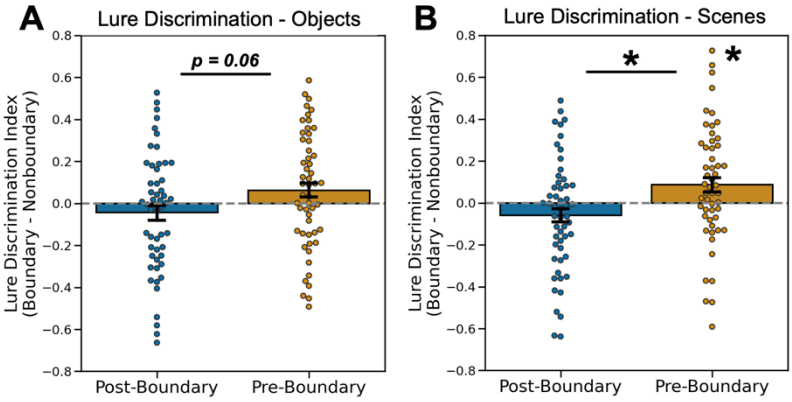
In the experiments reported in the main findings, we had no *a priori* hypotheses about whether the influence of event boundaries on later recognition memory would be influenced by the stimulus type. As such, our primary analyses collapsed across objects and scenes to maximize statistical power, given that a total of 24 event boundaries were present at encoding. Nonetheless, it is possible that either objects or scenes were especially well or poorly remembered, or that any effects due to event boundary adjacency interacted with stimulus type.

We repeated the Target recognition and Lure discrimination analyses reported in the main text (Fig. 2) separately for objects and scenes. For object trials, Target recognition accuracy was below baseline for Post-Boundary trials ($t_{(52)}$ = -3.38, p = 0.001) (Supplementary Fig. 1A). Post-Boundary Target recognition was marginally but not significantly worse than Pre-Boundary Target recognition ($t_{(104)}$ = -1.92, p = 0.060). For scene trials, Target recognition accuracy was below baseline for Post-Boundary trials ($t_{(52)}$ = -2.63, p = 0.011) (Supplementary Fig. 1B). Post-Boundary Target recognition was not significantly worse than Pre-Boundary Target recognition ($t_{(104)}$ = -1.01, p = 0.316). Thus, though the pairwise comparison between Pre-Boundary and Post-Boundary trials was not significant if analyzed separately, the data were very similar overall to the combined analyses.



Supplementary Figure 1: Experiment 1 Target recognition results for object and scene trails. Target recognition performance (d') as a function of adjacency to an event boundary during encoding is shown separately for object blocks (A) and scene blocks (B). (Asterisks indicate significant result at p < 0.05.)

We next turned to Lure discrimination. For object trials, Lure discrimination accuracy was not significantly above baseline for Pre-Boundary trials ($t_{(52)}$ = 1.64, p = 0.107) (Supplementary Fig. 2A). Pre-Boundary Lure discrimination was marginally but not significantly better than Post-Boundary Lure discrimination ($t_{(104)}$ = 1.91, p = 0.061). For scene trials, Lure discrimination accuracy was significantly above baseline for Pre-Boundary trials ($t_{(52)}$ = 2.25, p = 0.029) (Supplementary Fig. 2B). Furthermore, Pre-Boundary Lure discrimination was better than Post-Boundary Lure discrimination ($t_{(104)}$ = 2.82, p = 0.007). Like the Target recognition results above, the overall pattern of results was qualitatively similar between object and scene trials. However, the effects seemed to be somewhat stronger for Lure discrimination in scene blocks.

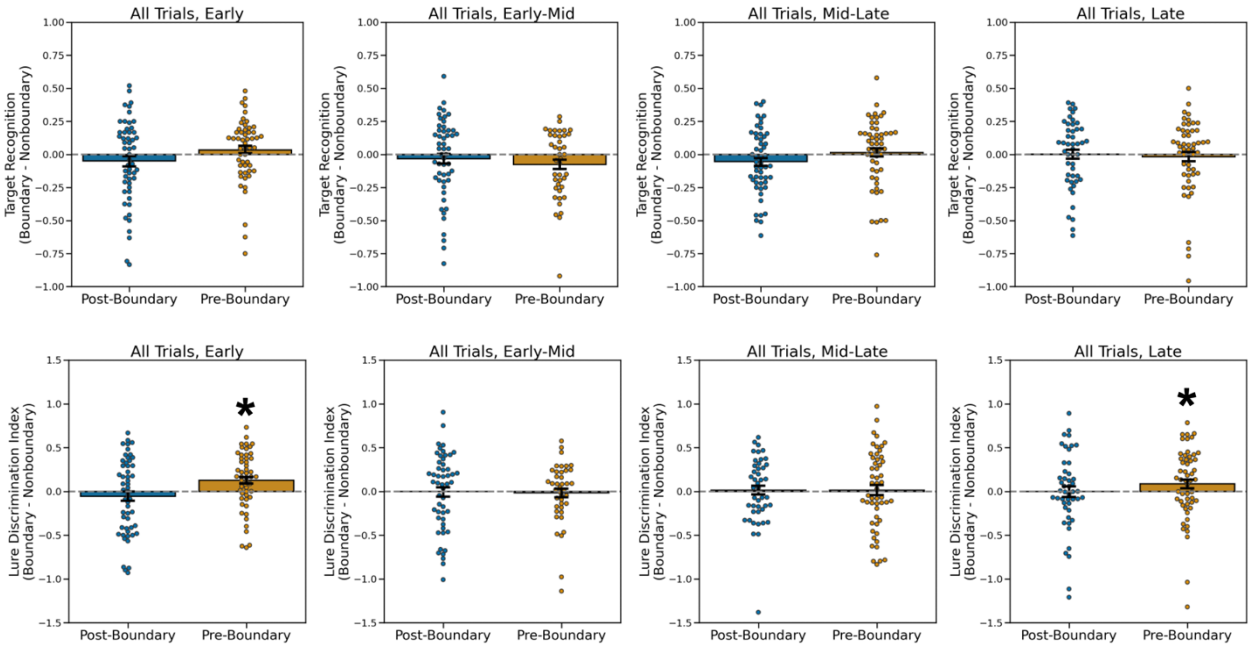


Supplementary Figure 2: Experiment 1 Lure discrimination results for object and scene trails. Lure discrimination performance (d') as a function of adjacency to an event boundary during encoding is shown separately for object blocks (A) and scene blocks (B). (Asterisks indicate significant result at p < 0.05.)

Experiment 1 analyses broken down by encoding quartile

As noted in the main text, we designed our experiments in such a way that encoding block length did not vary, but rather was held constant at 10 images. Thus, it is possible that the predictability of the block structure influenced the effects of event boundaries on recognition memory. Though experiments directly contrasting predictable versus unpredictable block lengths would be necessary to definitively answer that question, we reanalyzed the data here in such a way as to provide clues of such an influence. Specifically, we broke test trials down into four quartiles based on their temporal position at encoding. That is, we separately tested for Post-Boundary Target recognition deficits and Pre-Boundary Lure discrimination enhancements in each quartile. We caution that, given that this reduces each quartile to observing 6 event boundaries per participant at most (i.e., assuming all responses were made to those trials), this is an inherently noisy and underpowered analysis.

We reasoned that if learning to anticipate shifts in encoding blocks drove the observed effects, then we should see the effects selectively at either the first *or* the last quartile, but we should not see the effect in both or neither of those two quartiles. If the effects were found in the first but not the fourth quartile, it would suggest that they are driven by *unpredictability*. If the effects were found in the fourth quartile but not the first quartile, it would suggest that they are driven by *predictability*. As can be seen in Supplementary Figure 3, neither of these possibilities is obviously the case. For Targets (Supplementary Fig. 3, top row), we do not find any one quartile to carry significant effects of Post-Boundary recognition deficits alone. Furthermore, we find above-baseline Lure discrimination in the first ($t_{(52)} = 2.69$, $p = 0.009$) and fourth ($t_{(52)} = 2.11$, $p = 0.039$) quartiles. Thus, while we cannot absolutely rule out the possibility that block length predictability influenced our findings, the results here are not consistent with that possibility.



Supplementary Figure 3: Experiment 1 results broken down by four temporal quartiles during which the tested item was encountered during encoding. Top: Target recognition as a function of encoding quartile. Bottom: Lure discrimination as a function of encoding quartile. (Asterisks indicate significant result at $p < 0.05$.)